Tutorial - 7

① (a) In spherical coordinates:

$x = r\sin\theta\cos\phi$; $y = r\sin\theta\sin\phi$; $z = r\cos\theta$

$\cos\theta \propto Y_{10}$; $\sin\theta\cos\phi \propto (Y_{1-1} + Y_{1-1})$

$\sin\theta\sin\phi \propto (Y_{1-1} - Y_{1-1})$

$\therefore \psi(x,y,z) = r[\sin\theta\cos\phi + \sin\theta\sin\phi + 3\cos\theta]\, f(r)$

$= R(r) \times [a Y_{10} + b Y_{1-1} + c Y_{11}]$

$\therefore L^2 Y_{lm}(\theta,\phi) = \hbar^2 l(l+1) Y_{lm}(\theta,\phi) = 2\hbar^2 Y_{lm}(\theta,\phi)$.

$\therefore L^2 \psi(x,y,z) = R(r) \times [a Y_{10} + b Y_{1-1} + c Y_{11}]\, 2\hbar^2$

$$\boxed{L^2 \psi(x,y,z) = 2\hbar^2 \psi(x,y,z)}$$

$\therefore \psi(r)$ is an eigenfunc of L^2.

Eigenvalue is $2\hbar^2$.

(b) $L_z Y_{lm} = m\hbar Y_{lm}$

$\because l = 1$; possible values of m are $1, 0, -1$.

$\therefore \psi(r) = R(r)[\sin\theta\cos\phi + \sin\theta\sin\phi + 3\cos\theta]$.

$Y_{10} = \sqrt{\frac{3}{4\pi}}\cos\theta \Rightarrow \boxed{\cos\theta = \sqrt{\frac{4\pi}{3}}\, Y_{10}}$ — ①

$Y_{11} = \sqrt{\frac{3}{8\pi}}\sin\theta\, e^{i\phi}$; $Y_{1-1} = \sqrt{\frac{3}{8\pi}}\sin\theta\, e^{-i\phi}$

$Y_{11} + Y_{1-1} = \sqrt{\frac{3}{8\pi}}\sin\theta \cdot 2\cos\phi$

$\Rightarrow \boxed{\sin\theta\cos\phi = \sqrt{\frac{2\pi}{3}}(Y_{11} + Y_{1-1})}$ — ⑪

$\therefore$ ~~sinθ~~ $Y_{11} - Y_{1-1} = \sqrt{\frac{3}{8\pi}}\sin\theta \cdot 2i\sin\phi$

$\Rightarrow \boxed{\sin\theta\sin\phi = -i\sqrt{\frac{2\pi}{3}}(Y_{11} - Y_{1-1})}$ — ⑩

~~$\therefore \psi(r) = R(r)[a Y_{10} + b(Y_{11} + c Y_{1-1})]$~~

$\psi(r) = R(r)\left[\sqrt{\frac{2\pi}{3}}(Y_{11} + Y_{1-1}) - i\sqrt{\frac{2\pi}{3}}(Y_{11} - Y_{1-1}) + 3\sqrt{\frac{4\pi}{3}}Y_{10}\right]$

$\psi(r) = R(r)\left[Y_{11}\underbrace{(1-i)\sqrt{\frac{2\pi}{3}}}_{c_{11}} + Y_{1-1}\underbrace{\sqrt{\frac{2\pi}{3}}(1+i)}_{c_{1-1}} + \underbrace{\sqrt{\frac{4\pi}{3}}\cdot 3}_{c_{10}} Y_{10}\right]$

$\therefore$ ~~$P(m=1) = |c_{11}|^2$~~

$|c_{11}|^2 + |c_{1-1}|^2 + |c_{10}|^2$

$= \frac{4\pi}{3} + \frac{4\pi}{3} + \frac{36\pi}{3} = \frac{44\pi}{3} = \sum_{n=-1}^{1} |c_{1n}|^2$

$\therefore P(m=1) = \dfrac{|c_{11}|^2}{\sum |c_{1n}|^2} = \dfrac{\frac{4\pi}{3}}{\frac{44\pi}{3}} = \dfrac{1}{11}$ $[L_z Y_{11} = \hbar Y_{11}]$

$\therefore P(m=0) = \dfrac{|c_{10}|^2}{\sum |c_{1n}|^2} = \dfrac{36\pi/3}{44\pi/3} = \dfrac{9}{11}$ [Ans]

$[L_z Y_{10} = 0]$

$\therefore P(m=-1) = \dfrac{|c_{1-1}|^2}{\sum |c_{1n}|^2} = \dfrac{4\pi/3}{44\pi/3} = \dfrac{1}{11}$ $[L_z Y_{1-1} = -\hbar Y_{1-1}]$

Possible outcomes of L_z are $-\hbar, 0, \hbar$.
$\left(\frac{1}{11}\right) \left(\frac{9}{11}\right) \left(\frac{1}{11}\right)$.

(2) $H = \dfrac{J}{\hbar^2}(S_x^2 + S_y^2) + \dfrac{B}{\hbar} S_z$.

$$\boxed{H = \frac{J}{\hbar^2}(S^2 - S_z^2) + \frac{B}{\hbar} S_z}$$

$\therefore S^2 |s, m\rangle = \hbar^2 s(s+1) |s, m\rangle$.

Here $s = \frac{3}{2}$: $\therefore S^2 |\frac{3}{2}, m\rangle = \frac{15}{4}\hbar^2 |\frac{3}{2}, m\rangle$.

$S_z |\frac{3}{2}, m\rangle = m\hbar |\frac{3}{2}, m\rangle$.

~~$m = \frac{3}{2}$~~ :

~~$S^2 |\frac{3}{2}, \frac{3}{2}\rangle = \frac{15}{4}\hbar^2 |\frac{3}{2}, \frac{3}{2}\rangle$.~~

~~$S_z |\frac{3}{2}, \frac{3}{2}\rangle = \frac{3}{2}\hbar |\frac{3}{2}, \frac{3}{2}\rangle$.~~

~~$\therefore H |\frac{3}{2}, \frac{3}{2}\rangle =$~~

$H |\frac{3}{2}, m\rangle = \left[\frac{J}{\hbar^2}\left(\frac{15}{4}\hbar^2 - m^2\hbar^2\right) + \frac{B}{\hbar} \cdot m\hbar\right] |\frac{3}{2}, m\rangle$.

$$\boxed{E_m = \left[\left(\frac{15}{4} - m^2\right) J + Bm\right]}$$

	m	E_m
(i)	$\frac{3}{2}$	$\frac{3}{2}J + \frac{3}{2}B$.
(ii)	$\frac{1}{2}$	$\frac{7}{2}J + \frac{B}{2}$.
(iii)	$-\frac{1}{2}$	$\frac{7}{2}J - \frac{B}{2}$
(iv)	$-\frac{3}{2}$	$\frac{3}{2}J - \frac{3}{2}B$.

Eigenvectors are the states $|\frac{3}{2}, m\rangle$.

$m = \frac{3}{2}, \frac{1}{2}, -\frac{1}{2}, -\frac{3}{2}$

& eigenvalues are E_m.

(3) $L_x = \frac{1}{2}(L_+ + L_-)$

$L_\pm |l, m\rangle = \hbar\sqrt{l(l+1) - m(m\pm1)}\, |l, m\pm1\rangle$

$\therefore \langle L_\pm\rangle = \langle l, m| L_\pm |l, m\rangle = 0$ [as $|l, m+1\rangle$ & $|l, m-1\rangle$ are orthogonal to $|l, m\rangle$].

$\Rightarrow \langle L_x\rangle = \frac{1}{2}[\langle L_+\rangle + \langle L_-\rangle]$

$$\boxed{\langle L_x\rangle = 0}$$ (Ans)

$\therefore L_x^2 + L_y^2 + L_z^2 = L^2$

$$\Rightarrow \langle L_x^2 \rangle + \langle L_y^2 \rangle + \langle L_z^2 \rangle = \langle L^2 \rangle \qquad [\langle L_x^2 \rangle = \langle L_y^2 \rangle]$$

$$\Rightarrow \quad 2\langle L_x^2 \rangle = \langle L^2 \rangle - \langle L_z^2 \rangle$$

$$= \hbar^2 [l(l+1) - m^2]$$

$$\Rightarrow \boxed{\langle L_x^2 \rangle = \frac{\hbar^2}{2} [l(l+1) - m^2]} \quad \text{(Ans)}$$